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United States Patent	12385426
Kind Code	B2
Date of Patent	August 12, 2025
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Exhaust manifold

Abstract

An exhaust manifold includes first and second plate members fixed to an inlet of a junction pipe at which exhaust gases from exhaust pipes join together. The first and second plate members are combined with each other so as to partition the inlet for each of the exhaust pipes. The first plate member includes a slit into which the second plate member is inserted. A width of the slit is greater than a plate thickness of the second plate member. The first plate member is provided with first and second contact portions. The first and second contact portions respectively protrude from opposite sides of the slit in a width direction of the slit and contact with the second plate member inserted into the slit.

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Appl. No.:	18/479141
Filed:	October 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240175388 A1	May. 30, 2024

Foreign Application Priority Data

JP	2022-191687	Nov. 30, 2022
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Publication Classification

Int. Cl.: F01N13/10 (20100101); F01N13/18 (20100101)

U.S. Cl.:

CPC F01N13/10 (20130101); F01N13/1805 (20130101);

Field of Classification Search

CPC: F01N (13/10); F01N (13/1805); F01N (13/1872)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2022-191687, filed on Nov. 30, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an exhaust manifold.

BACKGROUND

(3) For example, Japanese Unexamined Patent Application Publication No. 2018-096347 discloses an exhaust manifold including a pair of plates joined to a fixing member that fixes a plurality of exhaust pipes collected on a downstream side of exhaust gas such that the plates cross each other in a cross shape via slits. Since the width of the slit is larger than the thickness of each plate so that the plates are easily coupled to each other, a predetermined clearance exists between the slit and the plate.

(4) Therefore, the plate might be thermally deformed in a direction in which the slit width is reduced. On the other hand, a circular region having a diameter longer than the slit width is formed

at an end portion opposite to an opening end of the slit so as to reduce concentration of stress due to, for example, thermal deformation.

(5) However, depending on the size of the circular region, the margin between the slits might become large. As a result, the positioning accuracy of each plate might decrease, and the concentration of stress on the slit might not be sufficiently reduced. When the stress is concentrated on the slit, a crack or the like might occur in the plate from the slit as a starting point. In this way, the reliability of the exhaust manifold might be reduced.

SUMMARY

(6) It is therefore an object of the present disclosure to provide an exhaust manifold with improved reliability.

(7) The above object is achieved by an exhaust manifold including first and second plate members fixed to an inlet of a junction pipe at which exhaust gases from exhaust pipes join together, wherein the first and second plate members are combined with each other so as to partition the inlet for each of the exhaust pipes, the first plate member includes a slit into which the second plate member is inserted, a width of the slit is greater than a plate thickness of the second plate member, the first plate member is provided with first and second contact portions, and the first and second contact portions respectively protrude from opposite sides of the slit in a width direction of the slit and contact with the second plate member inserted into the slit.

(8) The first and second contact portions may be provided at an opening end of the slit.

(9) The first and second contact portions may be provided so as to sandwich the second plate member in a thickness direction of the second plate member.

(10) Each of the first and second contact portions may be formed into a substantially trapezoidal shape in a front view of a plate surface of the first plate member, and the substantially trapezoidal shape includes an upper base that contacts with the second plate member.

(11) Also, the above object is achieved by an exhaust manifold including first and second plate members fixed to an inlet of a junction pipe at which exhaust gases from exhaust pipes join together, wherein the first and second plate members are combined with each other so as to partition the inlet for each of the exhaust pipes, the first plate member includes a slit into which the second plate member is inserted, a width of the slit is greater than a plate thickness of the second plate member, and the first plate member is welded to the second plate member in at least a part of the slit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating an example of an exhaust manifold;

(2) FIG. 2 is a perspective view illustrating an example of plate members before and after combination;

(3) FIG. 3 is a cross-sectional view of exhaust pipes and the plate members taken along line A-A in FIG. 1;

(4) FIG. 4 is a plan view illustrating an example of one of the plate members;

(5) FIG. 5 is a plan view illustrating the other of the plate members; and

(6) FIG. 6 is a perspective view illustrating an example of plate members that are welded to each other.

DETAILED DESCRIPTION

(7) FIG. 1 is a perspective view illustrating an example of an exhaust manifold 1. The exhaust manifold 1 includes exhaust pipes 11 to 14 and a collector 10. The exhaust pipes 11 to 14 extend from exhaust ports (not illustrated) of an internal combustion engine such as an engine. In the collector 10, as indicated by dotted arrows, the exhaust gases flowing through the exhaust pipes 11

to **14** join together. An exhaust purification device (not illustrated) for purifying exhaust gas, for example, is connected to the downstream side of the collector **10**. Although the number of cylinders of the internal combustion engine is four in the present embodiment, the number of cylinders is not limited.

(8) The collector **10** is an example of a junction pipe. The cross-sectional area of the flow path in the collector **10** is greater than the sum of the cross-sectional areas of the exhaust pipes **11** to **14**. Downstream end portions of the exhaust pipes **11** to **14** are gathered at an inlet **10a** of the collector **10** and connected to the collector **10** by welding or the like. In the lower part of FIG. **1** there is illustrated an enlarged view of the inlet **10a** of the collector **10**, indicated by reference numeral P. In the enlarged view, the exhaust pipe **12** is illustrated by a dotted line for convenience.

(9) A pair of plate members **21** and **22** combined such that their plate surfaces are substantially orthogonal to each other is fixed to the inlet **10a** by welding. Each of plate surfaces of the plate members **21** and **22** is formed into a substantially rectangular shape. When the inlet **10a** is viewed from the front, the plate members **21** and **22** are combined to form a cross shape. The plate members **21** and **22** are provided so as to partition the inlet **10a** for each of the exhaust pipes **11** to **14**.

(10) FIG. **2** is a perspective view illustrating an example of the plate members **21** and **22** before and after combination. The plate member **21** has a coupling slit **210**. The coupling slit **210** opens to the outer side of the collector **10** when viewed from the inlet **10a**. The plate member **22** has a coupling slit **220**. The coupling slit **220** opens to the inner side of the collector **10** when viewed from the inlet **10a**.

(11) The plate members **21** and **22** are combined with each other through the coupling slits **210** and **220**. Each width of the coupling slits **210** and **220** is greater than each plate thicknesses of the plate members **21** and **22**. The coupling slit **210** sandwiches a plate thickness portion on the extension line of the coupling slit **220**. The coupling slit **220** sandwiches a plate thickness portion on the extension line of the coupling slit **210**. The plate members **21** and **22** are fixed to the inlet **10a** of the collector **10** in a state where the coupling slits **210** and **220** are engaged with each other. As for the plate member **22**, the opening end of the coupling slit **220** is directed to the exhaust port side in FIG. **1**. In the plate member **21**, the opening end of the coupling slit **210** is directed to the exhaust gas purifying apparatus side in FIG. **1**.

(12) The plate members **21** and **22** are fixed to a seat plate **100** provided at a bottom portion of the inlet **10a** and having a cross shape in a front view. The seat plate **100** includes two metal plates **101** and **102** that are substantially orthogonal to each other. A bottom side portion **21D** along the long side of a plate surface **21s** of the plate member **21** is welded to the metallic plate **102**. A bottom side portion **22D** along a plate surface **22s** of the plate member **22** is welded to the metal plate **101**.

(13) FIG. **3** is a cross-sectional view of the exhaust pipes **11** to **14** and the plate members **21** and **22** taken along line A-A in FIG. **1**. FIG. **3** illustrates a front view of the inlet **10a** of the collector **10**. The plate members **21** and **22** partition the inlet **10a** into four sections for each of the exhaust pipes **11** to **14**. Side portions **21L**, **21R**, **22L**, **22R** at both ends in the long-side direction of the plate members **21** and **22** are welded to an inner peripheral surface **10s** of the collector **10**.

(14) Parts of outer peripheral surfaces **11a** to **14a** of the exhaust pipes **11** to **14** are in contact with the plate surfaces **21s** and **22s** of the plate members **21** and **22** and the inner peripheral surface **10s** of the collector **10**, and are fixed thereto by welding. Downstream end surfaces of the exhaust pipes **11** to **14** are welded to a bottom portion **10b** of the inlet **10a** so that flow paths **110** to **140** of the exhaust pipes **11** to **14** and a flow path (not illustrated) of the collector **10** communicate with each other in an airtight state.

(15) FIG. **4** is a plan view illustrating an example of the plate member **21**. FIG. **4** illustrates the plate member **21** when the plate surface **21s** is viewed from the front. The coupling slit **210** of the plate member **21** opens on an upper side portion **21U**. The plate member **22** is inserted into the coupling slit **210**. Here, a part of the plate member **22** is indicated by a dotted line. The coupling

slit **210** is an example of a slit.

(16) High-temperature exhaust gas discharged from the internal combustion engine flows through the exhaust pipes **11** to **14**. Therefore, thermal expansion occurs in the plate members **21** and **22**. In contrast, the atmosphere exists around the collector **10**. Therefore, the collector **10** does not thermally expand. Therefore, a stress F_a acts on the plate member **21** in a direction in which the width of the coupling slit **210** is narrowed.

(17) When the plate member **21** is deformed such that the width of the coupling slit **210** is narrowed by the concentrated action of the stress F_a , a crack or the like might occur in the plate member **21** from the coupling slit **210** as a starting point. As a result, the reliability of the exhaust manifold **1** might be reduced. Thus, a circular region r is formed in an end portion **210b** so as to reduce the concentration of the stress F_a . The circular region r has a diameter greater than the width h of the coupling slit **210**.

(18) However, depending on the size of the circular region r , the margin between the coupling slits **210** and **220** might become excessively large, the positioning accuracy of the plate members **21** and **22** might be lowered, and the concentration of the stress F_a on the coupling slit **210** might not be sufficiently reduced.

(19) Therefore, the plate member **21** is provided with contact portions **211** and **212** which protrude from opposite sides of the coupling slit **210** in the width direction thereof and contact with the inserted plate member **22**. For this reason, even if the stress F_a acts on the coupling slit **210**, since the contact portions **211** and **212** are in contact with the plate member **22**, the width of the plate member **21** is suppressed from being deformed to narrow.

(20) Reference numeral M indicates an enlarged view of the vicinity of the contact portions **211** and **212**. The contact portions **211** and **212** are provided at an opening end **210a** of the coupling slit **210**. Therefore, as compared with a case where the contact portions **211** and **212** are provided on the end portion **210b** opposite to the opening end **210a**, the distance from the root of the coupling slit **210** is increased. Therefore, the deformation of the plate member **21** due to the stress F_a is effectively suppressed. The circular region r is not always required.

(21) The contact portions **211** and **212** are provided so as to sandwich the plate member **22** in the thickness direction. That is, the contact portions **211** and **212** are provided at positions opposed to each other across the coupling slit **210**. Therefore, as compared with the case where the contact portions **211** and **212** are provided at different positions in the extending direction of the coupling slit **210** so as not to face each other, the stress F_a from the left and right sides of the coupling slit **210** is offset across the plate member **22**. Therefore, the deformation of the plate member **21** due to the stress F_a is effectively suppressed.

(22) Each of the contact portions **211** and **212** has a substantially rectangular shape in a front view of the plate surface **21s**. The contact portions **211** and **212** have the upper bases of the substantially trapezoidal shapes in contact with the plate member **22**. For this reason, compared with, for example, a case where the contact portions **211** and **212** are substantially rectangular, it is easier to concentrate the stress F_a on the plate member **22**. Therefore, the deformation of the plate member **21** due to the stress F_a is effectively suppressed. The length of the lower base of the substantially trapezoidal shape is shorter than the length of the coupling slit **210**.

(23) Widths d (heights of the substantially trapezoidal shapes) of the contact portions **211** and **212** are substantially equal to each other, and are smaller than a width h of the coupling slit **210**. The sum of the widths d of the contact portions **211** and **212** and the plate thickness t of the plate member **22** is substantially equal to the width h . Therefore, the plate member **22** is press-fitted into the coupling slit **210** of the plate member **21**. The width d of each of the contact portions **211** and **212** is preferably set to, for example, one half of the plate thickness t from the viewpoint of suppressing deformation.

(24) FIG. 5 is a plan view illustrating the plate member **22**. FIG. 5 illustrates the plate member **22** when the plate surface **22s** is viewed from the front. The coupling slit **220** of the plate member **22**

opens on the side of the bottom side portion 22D. The plate member 21 is inserted into the coupling slit 220. The coupling slit 220 is engaged with the plate member 21 through the coupling slit 210. Here, a part of the plate member 21 is indicated by a dotted line. The coupling slit 220 is an example of a slit.

(25) Similarly to the plate member 21 described above, the stress F_b acts on the plate member 22 in the direction of narrowing the width of the coupling slit 220. Thus, a circular region R is formed in an end portion 220b so as to reduce the concentration of the stress F_b . The diameter of the circular region R is greater than the width h of the coupling slit 220. However, there might be a case where the concentration of the stress F_b is not sufficiently reduced for the reason described above.

(26) Therefore, the plate member 22 is also provided with contact portions 221 and 222, similar to the contact portions 211 and 212 described above. The contact portions 221 and 222 protrude from both sides in the width direction of the coupling slit 220 and contact with the inserted plate member 21. Therefore, even when the stress F_b is applied to the coupling slit 220, the contact portions 221 and 222 are in contact with the plate member 21. For this reason, the width of the plate member 22 is suppressed from being deformed to narrow. The circular region R is not always required.

(27) Each of the contact portions 221 and 222 has a substantially trapezoidal shape in a front view of the plate surface 22s. The contact portions 221 and 222 are provided on an opening end 220a of the coupling slit 220 so as to sandwich the plate member 21 in the thickness direction thereof. Therefore, similarly to the contact portions 211 and 212 described above, it is possible to effectively suppress deformation of the plate member 22 due to the stress F_b .

(28) In the present embodiment, the deformation of the plate members 21 and 22 is suppressed by the contact portions 211, 212, 221, and 222, but the present disclosure is not limited thereto. The deformation may be suppressed by welding the plate members 21 and 22 to each other as follows.

(29) FIG. 6 is a perspective view illustrating an example of plate members 21a and 22a that are welded. In FIG. 6, the same components as those in FIG. 2 are denoted by the same reference numerals, and description thereof is omitted. Unlike the above-described plate members 21 and 22, the plate members 21a and 22a have coupling slits 210 and 220 in which the contact portions 211, 212, 221, and 222 are not formed.

(30) The plate member 21a is welded to the plate member 22a at the opening end 210a of the coupling slit 210. The opening end 210a is melted by welding to be a welded portion 3 having a substantially spherical shape, and the welded portion 3 is integrated with the plate thickness portion of plate member 22a. Note that the welded portion 3 is indicated by a dotted line for convenience.

(31) The deformation of the plate member 21a due to the stress F_a is suppressed by the welded portion 3. The welded portion 3 is not limited to the opening end 210a of the coupling slit 210, and may be another portion. That is, the plate member 21a may be welded to the plate member 22a in at least a part of the coupling slit 210. Also, the plate member 22a may be welded to the plate member 21a at least at a part of the coupling slit 220.

(32) As described above, since the deformation of the plate members 21, 21a, 22, and 22a is suppressed, the reliability of the exhaust manifold 1 is improved.

(33) Although some embodiments of the present disclosure have been described in detail, the present disclosure is not limited to the specific embodiments but may be varied or changed within the scope of the present disclosure as claimed.

Claims

1. An exhaust manifold comprising first and second plate members fixed to an inlet of a junction pipe at which exhaust gases from exhaust pipes join together, wherein the first and second plate members are combined with each other so as to partition the inlet for each of the exhaust pipes, the first plate member includes a slit into which the second plate member is inserted, the slit having an elongated shape, a width of the slit is greater than a plate thickness of the second plate member, the

first plate member is provided with first and second contact portions at an opening end of the slit, the first and second contact portions respectively protrude from opposite sides of the slit in a width direction of the slit and contact with the second plate member inserted into the slit, and a length of each of the first and second contact portions is shorter than a half of a length of the slit in a longitudinal direction of the elongated shape.

2. The exhaust manifold according to claim 1, wherein the first and second contact portions are provided so as to sandwich the second plate member in a thickness direction of the second plate member.

3. The exhaust manifold according to claim 1, wherein each of the first and second contact portions is formed into a substantially trapezoidal shape in a front view of a plate surface of the first plate member, and the substantially trapezoidal shape includes an upper base that contacts with the second plate member.
